

UNIT -II

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Mathematical models for decision making:

Structure of mathematical models, Development of a model, Classes of models
Data mining: Definition of data mining, Representation of input data, Data mining process, Analysis methodologies
Data preparation: Data validation, Data transformation, Data reduction

Structure of mathematical models:

Mathematical models are used in many fields like physics, engineering, and economics. These models have different structures but share common features.

A **model** is a simplified version of a real system, focusing only on important parts for study. It helps in understanding how a system works without including unnecessary details.

Example:

- A **map** is a model of a city. It shows roads and landmarks but not every building or tree.
- A **weather forecast model** predicts rain based on temperature, humidity, and wind, ignoring unnecessary details like street names.

Einstein said, "**Make it as simple as possible, but not simpler.**" This means a model should be detailed enough to be useful but not overly complicated.

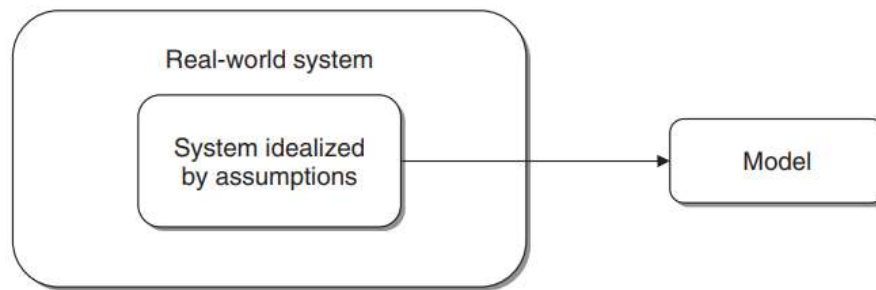


Figure 4.1 A model is a selective abstraction of reality

Mathematical Models:

1. Use of Mathematical Models:
 - Scientific and technological advancements rely on mathematical models to represent real-world systems.
 - These models help analyze and predict behaviors using abstract methods.
2. Thought Experiment (Gedankenexperiment):
 - A mental model is created to study a phenomenon by imagining the effects of changes.
 - Example: A company can use a spreadsheet for a "what-if" analysis to see how changing invoice payment terms (90 days vs. 60 days) affects cash flow.
3. Types of Models:
 - Iconic Models: Physical miniatures of real-world systems.
 - Example: A small-scale model of a city neighborhood.
 - Analogical Models: Physical models that imitate behavior by analogy.
 - Example: A wind tunnel simulating a car's aerodynamics.
 - Symbolic Models: Abstract models using mathematical formulas and symbols.
 - Example: Business Intelligence (BI) models based on data analysis.
4. Models Based on Probability:
 - Stochastic Models: Include randomness and probability distribution.
 - Example: Predictive models and waiting line models.
 - Deterministic Models: All inputs are known in advance.
 - Example: Used when randomness is minimal, like in supply chain planning.
5. Models Based on Time:
 - Static Models: Analyze a system at a single point in time.
 - Example: Optimizing the distribution of goods for a specific period.

- **Dynamic Models:** Study changes over multiple time periods.
- **Discrete-Time Models:** Observe changes at fixed intervals (minutes, days, years).
- **Continuous-Time Models:** Track changes continuously over time.

Mathematical models simplify complex systems, helping in decision-making across various fields like business, science, and engineering.

Development of a model:

Phases of Developing a Mathematical Model for Decision Making

Developing a mathematical model involves four key phases, including a feedback mechanism for revisions.

1. Problem Identification

- The first step is identifying the problem clearly.
- Symptoms of the issue must be analyzed and understood.
- **Example:** A company has a high stock level and excessive stock turnover. The cause may be poor production planning.

2. Model Formulation

After identifying the problem, a suitable mathematical model must be developed.

- **Time Horizon:**
 - Defines the time frame for decision-making.
 - **Example:** Weekly production plans for a year (long-term) vs. daily task assignments (short-term).
- **Evaluation Criteria:**
 - Used to measure and compare different decisions.
- **Example Factors:**

- **Cost & Profits:** Monetary benefits or expenses.
- **Service Effectiveness:** How well the system operates.
- **Quality:** Product or service standards.
- **Flexibility:** Ability to adapt to changes.
- **Reliability:** Meeting targets consistently.
- **Decision Variables:**
 - Define symbolic variables that represent choices.
 - **Example:** Production volume for each product, process, and time period.
- **Numerical Parameters:**
 - Essential values needed in calculations.
 - **Example:** Available factory capacity and production efficiency.
- **Mathematical Relationships:**
 - Equations connecting decision variables, parameters, and performance indicators.
 - **Example:** A model may use probability to account for uncertain demand.
 - A balance between accuracy and complexity is essential for practical solutions.

3. Development of Algorithms

- After creating the model, it must be solved using algorithms.
- A software tool is developed or acquired for processing.
- The analyst must understand different solution methods.

4. Implementation & Testing

- The model is implemented and tested before use.
 - Data accuracy from data warehouses must be verified.
 - Validation is done by decision-makers to check:
 - **Realistic Results:** Are the conclusions logical?
 - **Extreme Value Testing:** Do results remain consistent at maximum/minimum values?
 - **Stability Check:** Are results reliable when minor inputs change?
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Class model:

Mathematical models help in making better decisions. Different types of models are used for different decision-making processes. Each model type has specific techniques to solve problems.

- Predictive models
- Pattern recognition and learning models
- Optimization models
- Project management models
- risk analysis models
- waiting line models.

A. Predictive Models :

Predictive models are crucial in business intelligence as they help estimate future events based on past data. These models provide important inputs for decision-making in various business areas.

1. Importance of Predictive Models

- Many business intelligence models, such as **optimization models**, require future data for decision-making.
- Predictions help businesses plan for **demand, prices, costs, and technology trends**.
- Since predictions come before decision-making, predictive models are a **starting point** in business intelligence.

2. Role in Business Decisions

Predictive models are used in different departments of a company:

- **Strategy & Research:** Predicts market trends and future opportunities.
- **Administration & Control:** Estimates financial risks and revenue forecasts.
- **Marketing:** Analyzes customer behavior to improve sales strategies.
- **Production & Logistics:** Forecasts product demand and supply chain needs.

3. Types of Predictive Models

A. Explanatory Models

- Identify relationships between a **dependent variable** (outcome) and **independent variables** (factors).
- Used in **regression analysis** and **classification models**.
- **Example:** A company predicts sales based on advertising budget and customer income levels.

B. Time Series Models

- Analyze patterns in **historical data** over time to make future predictions.
- **Example:** A retail store forecasts monthly sales based on past sales data

B. Pattern recognition and machine learning models

Pattern recognition and learning help machines extract knowledge from past experiences and apply it in the future. This concept is inspired by human intelligence and is used to create **intelligent systems** that can learn and improve over time.

1. Purpose of Pattern Recognition and Learning

- Helps understand how intelligence develops by learning from past data.

- Uses **mathematical models** to create efficient algorithms for machines to learn.
- Enables **AI systems** to analyze past information and make future decisions, just like humans.

2. Applications of Learning Models

Mathematical learning models are used in many fields, such as:

- **Image, Sound, and Text Recognition:** Face recognition, speech processing, and handwriting recognition.
- **Medical Diagnosis:** Identifying diseases based on symptoms and medical history.
- **Marketing:** Customer segmentation and behavior analysis.
- **Manufacturing Control:** Detecting faults in production processes.
- **Fraud Detection:** Identifying unusual patterns in transactions.

3. Types of Learning Models

A. Interpretation Models

- Identify patterns in data and express them in understandable rules.
- **Example:** A spam filter learns to recognize spam emails based on past patterns.

B. Prediction Models

- Forecast future values based on past data.
- **Example:** A weather app predicts tomorrow's temperature using past weather data.

4. Supervised vs. Unsupervised Learning

A. Supervised Learning

- The model learns from data with **labeled outputs** (a target variable).
- **Examples:**

- **Classification:** Sorting emails into "spam" and "not spam."
- **Regression:** Predicting house prices based on size and location.

B. Unsupervised Learning

- No labeled output; the model identifies patterns on its own.
- **Examples:**
 - **Clustering:** Grouping customers based on buying habits.
 - **Association Rules:** Finding items often bought together in a supermarket (e.g., bread & butter).

C. Optimization Models:

Optimization models are used by companies to make the best decisions by choosing from a set of possible options based on specific criteria, like cost or profit.

1. Decision-Making Process

- **Problem Definition:** Identify the problem and options available.
- **Evaluation Criteria:** Set a way to evaluate options, like cost or profit.
- **Optimal Decision:** Choose the best option that meets the goal (minimum cost or maximum profit).

Example: A company wants to minimize shipping costs. It compares different delivery options to find the cheapest choice.

2. Resource Allocation

Optimization models help allocate limited resources effectively. Resources could be **money, materials, people, or time**.

Examples of application areas:

- **Logistics & Production Planning:** Deciding how to use factory resources efficiently.
- **Financial Planning:** Allocating a budget for different projects.
- **Work Shift Planning:** Scheduling employees efficiently.

- **Marketing Campaigns:** Deciding where to spend advertising money.
- **Price Determination:** Setting prices to maximize profit.

3. Types of Optimization Models

Optimization models can take different forms based on the problem:

- **Linear Optimization:** Deals with linear relationships between decision variables.
- **Integer Optimization:** Decision variables must be whole numbers (e.g., number of items).
- **Convex Optimization:** Solves problems where the objective function has a specific curve shape.
- **Network Optimization:** Solves problems in network systems (e.g., transportation routes).
- **Multiple-Objective Optimization:** Deals with problems involving more than one goal (e.g., minimizing cost while maximizing quality).

D. Project Management Models :

A **project** is a collection of related tasks aimed at achieving a specific goal, such as building a plant, developing a product, or creating an information system. Successful project execution requires careful planning and control of tasks, resources (human, technical, and financial), and timelines.

1. Project Management Process

- **Planning & Control:** Involves organizing tasks and managing resources to achieve the project goal.
- **Interdependent Activities:** Tasks in a project depend on each other, requiring careful coordination.
- **Resources:** Includes people, materials, money, and technology needed to complete tasks.

2. Mathematical Models in Project Management

Mathematical models help manage projects efficiently by representing tasks and their relationships.

- **Network Models:**
 - Used to map tasks in a project and show their order (which tasks must happen before others).
 - Helps calculate the **overall project duration**.
 - **Example:** In constructing a building, certain tasks (like laying the foundation) must happen before others (like constructing walls).
- **Stochastic Models (PERT):**
 - Used when task durations are uncertain and have random variables (like weather delays).
 - Provides a **range of possible project completion times**.
 - **Example:** If a construction task may take between 5 to 7 days depending on weather, a stochastic model predicts the possible project completion time.
- **Optimization Models:**
 - Used to allocate limited resources efficiently to minimize costs or time.
 - Helps find the **best way to assign resources** to tasks while balancing costs and project timelines.
 - **Example:** Deciding how many workers to assign to each task to complete the project in the shortest time and at the lowest cost.

E. Risk analysis models :

Some decisions must be made without knowing future outcomes with certainty. Risk analysis models help in such cases by estimating possible future events and their impact.

1. Decision-Making with Uncertainty

- The decision-maker must choose among different options without knowing future effects.
- **Example:** A company deciding how to increase production capacity:
 - **Build a new plant** → High investment, low operating cost.
 - **Expand an existing plant** → Lower investment, higher operating cost.

- **Subcontract production** → Lowest investment, highest operating cost.

2. Impact of Future Events

- The best option depends on **future demand**, which is uncertain.
- **High demand** → Building a new plant is best.
- **Low demand** → Subcontracting is better.
- **Medium demand** → Expanding the existing plant is a good choice.
- Since demand is unknown, market research can **estimate probabilities** of different demand levels.

3. Risk Analysis Models

- Help in **evaluating risks** before making a decision.
- Based on **Bayesian and utility theories** to estimate probabilities and outcomes.
- Used in various fields, such as:
 - **Technology Investment** – Choosing new technologies.
 - **Product Design** – Assessing market acceptance.
 - **R&D Projects** – Evaluating potential success.
 - **Financial & Real Estate Investment** – Predicting risks and returns.

Conclusion

Risk analysis models help decision-makers **assess uncertainty, estimate future risks, and make informed choices** in business and investments.

F. Waiting line models :

- **Waiting Line Theory (Queueing Theory)**
- Waiting line theory studies congestion when **customer demand and service availability** are unpredictable. When many customers need a service at the same time, some must **wait in line**.

1. Components of a Waiting Line System

- **Source (Population):** Customers who need the service (finite or infinite).
- **Arrival Process:** Customers arrive randomly at different times.
- **Service Process:** Providers serve customers based on availability.
- **Waiting Area:** Customers wait if the service is not immediately available.
- **Service Stations:** Places where service is given (e.g., cash counters in a bank).
- **Waiting Rules:** Defines the order of service (e.g., first-come, first-served).
- **Priority System:** Some customers may get priority (e.g., emergency patients in a hospital).

2. Importance of Waiting Line Models

- Used to **design efficient service systems** by balancing:
 - **Service cost:** More staff reduces waiting but increases expenses.
 - **Waiting cost:** Longer wait times lead to unhappy customers.
- The goal is to find an **optimal balance** between service quality and cost.

3. Examples of Waiting Line Systems

- **Banks:** Customers wait for tellers.
- **Hospitals:** Patients wait for doctors.
- **Supermarkets:** Shoppers wait at billing counters.
- **Call Centers:** Customers wait for an agent to answer.

Conclusion

Waiting line models help businesses **reduce customer wait times, improve service efficiency, and control costs** by designing better queue management systems.

Data Mining :

Introduction to Data Mining

With advancements in technology, vast amounts of data are available in business, science, medicine, and public administration. **Data mining** is the process of analyzing large datasets to find useful patterns and support decision-making.

1. What is Data Mining?

- It is the **exploration and analysis** of large datasets.
- Helps in **finding patterns, extracting knowledge, and discovering rules**.
- Used in **decision-making** across various fields.

2. Related Concepts

- **Knowledge Discovery:** Finding valuable insights from data.
- **Pattern Recognition:** Identifying trends in data.
- **Machine Learning:** Using algorithms to improve predictions and automate decisions.

3. Importance of Data Mining

- Helps businesses improve strategies.
- Used in medical research for disease prediction.
- Assists in fraud detection and risk analysis.

4. Data Mining Process

- **Understanding the Data:** Identify key properties of the dataset.
- **Choosing Analysis Methods:** Select the right techniques for pattern discovery.
- **Extracting Knowledge:** Apply models to find useful insights.

Conclusion

Data mining is a powerful tool for extracting valuable information from large datasets, helping businesses, scientists, and governments make better decisions.

Definition of data mining :

Data Mining Process and Its Applications

Data mining is an **iterative process** used to analyze large databases and extract useful knowledge for decision-making. It involves **data collection, analysis, pattern recognition, and predictive modeling**.

1. Characteristics of Data Mining

- It requires **feedback and revisions** throughout different phases.
- It **cannot be fully automated** and needs expert intervention.
- Extracted knowledge must be **accurate and data-driven** to avoid misleading conclusions.

2. Steps in Data Mining

- **Data Gathering:** Collects information from various sources (e.g., sales records, customer logs).
- **Inductive Learning:** Uses mathematical models to **derive patterns and rules** from past data.
- **Decision-Making:** Applies discovered insights to take **practical actions**.

3. Data Mining vs. Classical Statistics

- Unlike traditional data collection, data mining **uses existing records**, not pre-planned samples.
- Example: A **retail store** records purchase history for billing but later uses it to analyze shopping trends.

4. Two Main Approaches in Data Mining

A. Interpretation (Pattern Discovery)

- Finds **hidden relationships** in data.
- **Example:** A retail store clusters **loyalty card holders** into groups based on buying habits to design better marketing strategies.

B. Prediction (Future Forecasting)

- Estimates future outcomes based on past data.
- **Example 1:** A **mobile company** predicts which customers may switch to another provider.
- **Example 2:** A **retail business** forecasts product sales for the next few weeks.

5. Dual-Purpose Models

- Some models work for **both prediction and interpretation**.
- **Example:** A **classification tree** used for prediction can also help **understand customer behavior**.

Interpretation vs. Prediction in Data Mining

Data mining helps in **understanding patterns** (Interpretation) and **forecasting future events** (Prediction).

1. Interpretation (Pattern Discovery)

- Identifies **regular patterns** in data and expresses them through rules.
- The rules must be **useful and non-trivial** to improve knowledge.
- **Example:**
 - A retail company **groups loyalty card customers** based on shopping habits.
 - Helps in **finding market niches** and improving **marketing strategies**.

2. Prediction (Future Forecasting)

- Estimates **future values or events** based on past data.
- **Example:**
 - A **mobile company** predicts which customers may **switch to a competitor**.
 - A **retail store** forecasts **product sales** for the next few weeks.
- Uses **data variables** like age, contract length, and calling patterns to make predictions.
- **Time series models** predict based only on past values of the same variable.

3. Overlap Between Interpretation and Prediction

- Some models help in **both** pattern recognition and forecasting.
- **Example:**
 - **Classification trees** predict customer behavior and also explain why patterns exist.

Conclusion

- **Interpretation** helps in understanding trends and improving decision-making.
- **Prediction** helps in forecasting future outcomes and preparing strategies.

Models and methods for data mining :

Learning Methods in Data Mining

Different learning methods help in **data mining tasks** like **pattern recognition and prediction**. These methods come from **computer science, statistics, and mathematics**.

1. Types of Learning Methods

- **Machine Learning (Computer Science):**
 - Uses **classification trees and association rules**.
 - Focuses on **discovering patterns from data**.
 - **Example:** A **spam filter** classifies emails as spam or not.
- **Statistical Methods (Mathematics & Statistics):**
 - Uses **regression and Bayesian classifiers**.
 - Based on **mathematical formulas**.
 - **Example:** Predicting **house prices** using regression analysis.
- **Statistical Learning Theory (Advanced Mathematics):**
 - Combines **probability, optimization, and statistics**.
 - Used for **highly accurate predictions**.
 - **Example:** AI-powered **stock price forecasting**.

2. Steps in Developing a Data Mining Model

1. **Select a Model Type:** Choose the best method to learn from past data.
2. **Define Evaluation Metrics:** Measure accuracy and effectiveness.
3. **Design an Algorithm:** Optimize the model for better predictions.

3. Application of Learning Methods

- Some methods can be used for **multiple data mining tasks** like classification and prediction.
- Each technique is linked to the **most suitable task** for better results.

Conclusion

Different learning methods help in **extracting patterns, making predictions, and improving decision-making** in data mining.

Data mining, classical statistics and OLAP :

How Data Mining Differs from Statistics and OLAP

Data mining is **different** from **classical statistics** and **OLAP (Online Analytical Processing)** in several ways.

1. Passive vs. Active Analysis

- **Statistics & OLAP (Passive Approach):**
 - Decision-makers **formulate a hypothesis first** and then test it.
 - **Example:** A company assumes **customer age affects sales** and checks past data to confirm.
 - Uses **top-down analysis** (starting with a question).
- **Data Mining (Active Approach):**
 - **Automatically discovers patterns** from data without predefined assumptions.
 - **Example:** A data mining model finds that **weekend sales are highest**, even without prior intuition.
 - Uses **bottom-up analysis** (starting with data insights).

2. Importance of Data Mining in Large Datasets

- When dealing with **huge data volumes**, it is difficult for humans to **form hypotheses manually**.
- **Data mining models** play a key role in generating **new knowledge** automatically.

Conclusion

Unlike **statistics and OLAP**, which confirm existing ideas, **data mining actively finds hidden patterns and makes predictions**, making it essential for handling big data.

Differences Between OLAP, Statistics, and Data Mining:

Feature	OLAP (Online Analytical Processing)	Statistics	Data Mining
Approach	Top-down (User asks specific questions)	Top-down (User tests hypotheses)	Bottom-up (Finds hidden patterns)
Purpose	Summarizing and visualizing data	Validating hypotheses	Discovering patterns and predicting outcomes
Process	Querying and analyzing structured data	Applying mathematical models	Using algorithms to find insights automatically
User Input	Requires user to define queries and reports	Requires user to define a hypothesis	Works independently to detect trends
Automation	Low – Users navigate data cubes	Medium – Statistical tests require manual setup	High – AI/ML algorithms explore data
Example Use Case	Sales dashboard showing monthly revenue	Testing if ad spending affects sales	Finding hidden customer behavior patterns
Best For	Business intelligence, reporting	Research, hypothesis testing	Predictive analytics, AI-driven insights

Applications of Data Mining

Data mining methodologies are used in various fields to extract useful insights and make decisions. Here are some common applications:

1. Relational Marketing

- **Customer Segmentation:** Identify customers who respond well to targeted campaigns (e.g., cross-selling).
- **Retention Campaigns:** Find customers likely to stay and engage with offers.
- **Response Prediction:** Estimate how many people will respond to marketing efforts.
- **Market Basket Analysis:** Understand which products are commonly bought together.

Example: A retailer uses data mining to target customers for specific promotions based on their buying patterns.

2. Fraud Detection

- **Financial Fraud:** Detect illegal activities like credit card theft, false claims, and unauthorized transactions.
- **Industries Affected:** Banking, insurance, and telecommunication.

Example: A bank uses data mining to identify unusual spending patterns that might suggest fraudulent transactions.

3. Risk Evaluation

- **Predictive Modeling:** Assess risks in future decisions, such as whether a bank should approve a loan.

Example: A bank predicts the likelihood of loan repayment based on the applicant's history and characteristics.

4. Text Mining

- **Text Classification:** Sort and categorize unstructured data such as emails, documents, and web pages.

- **Filters and Analysis:** Automatically filter emails or analyze press releases.

Example: A web search engine uses text mining to categorize and rank search results.

5. Image Recognition

- **Image Classification:** Recognize and classify digital images for different uses.
- **Applications:** Facial recognition, character recognition, and security surveillance.

Example: A security system uses data mining to detect suspicious activities in surveillance videos.

6. Web Mining

- **Clickstream Analysis:** Track user behavior on websites to improve user experience.
- **E-commerce Optimization:** Analyze website data to customize shopping experiences.

Example: An e-commerce site uses web mining to suggest products based on browsing history.

7. Medical Diagnosis

- **Disease Detection:** Use data mining to predict and diagnose diseases based on test results and patient history.
- **Image Analysis for Diagnosis:** Analyze medical images for early detection of health issues.

Example: A hospital uses data mining to predict the likelihood of a patient developing diabetes based on health data.

Conclusion

Data mining helps in **customer targeting, fraud detection, risk analysis, medical diagnosis, and more** by analyzing large sets of data to uncover useful patterns and insights.

Representation of input data :

Data Mining Dataset Structure

In data mining, the input data is typically organized into a **two-dimensional table** called a **dataset**, where:

- **Rows** represent **observations** (also called records or cases).
- **Columns** represent **attributes** (also called variables or features).

1. Types of Attributes

A. Categorical Attributes

- Have a **limited number of distinct values** (usually less than 100).
- Represent **qualitative data** (cannot be used in arithmetic operations).
- **Example:**
 - Province of residence (e.g., "New York", "California").
 - Customer status (e.g., 1 for leaving, 0 for staying).

B. Numerical Attributes

- Represent **quantitative data** that can be used in arithmetic operations like subtraction or division.
- **Example:**
 - Total amount spent by a customer.
 - Difference in spending between two customers (\$27 vs. \$36).

2. Types of Categorical Attributes

A. Counts

- Can only be **true or false** (binary).
- **Example:** A customer may or may not have a credit card.

B. Nominal

- **No natural order** in the values.
- **Example:** Province of residence (no ranking between different provinces).

C. Ordinal

- **Has a natural order**, but no arithmetic calculations can be made.
- **Example:** Education level (e.g., high school, college, graduate).

3. Types of Numerical Attributes

A. Discrete

- **Limited or countable** values.
- **Example:** Number of children in a family.

B. Continuous

- **Unlimited values** (can take any value within a range).
- **Example:** Monthly income, temperature.

4. Dataset Representation

- A dataset DD with **m observations** (rows) and **n attributes** (columns) is represented by a matrix XX of size **$m \times n$** .
 - **x_i** : An **n -dimensional row vector** for the i th record.
 - **a_j** : An **m -dimensional column vector** for the j th attribute.

Conclusion

In data mining, data is stored in a **two-dimensional dataset**, with **rows** for observations and **columns** for attributes, which can be **categorical or numerical**. The data is then analyzed for patterns or predictions.

Data mining process :

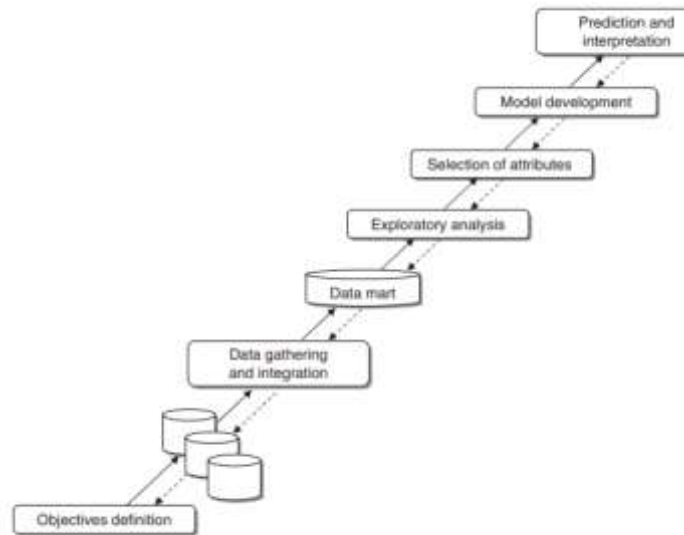


Figure 5.1 Data mining process

Phases of the Data Mining Process

Data mining is an **iterative process** that involves several key steps to extract useful knowledge from data. Here are the main phases:

1. Definition of Objectives

- **Goal:** Clearly define what you want to achieve with data mining.
- **Example:** Predict customer churn (who might leave the service) based on customer characteristics.
- **Collaboration:** Domain experts and data analysts work together to set clear objectives.

2. Data Gathering and Integration

- **Goal:** Collect relevant data from different sources (internal, external).
- **Challenges:** Data may need integration and can be in different formats.
- **Example:** Integrating customer data from both online and offline sources.
- **Risk:** Poorly structured data (e.g., only storing total amounts without item details) can harm analysis.

3. Exploratory Analysis

- **Goal:** Understand the data and clean it for analysis.
- **Tasks:** Check for errors, missing values, and outliers.
- **Example:** Correcting invalid birth dates or negative sales amounts in the data.
- **Tools:** Histograms and summary statistics help in identifying issues.

4. Attribute Selection

- **Goal:** Identify and keep only the relevant data attributes.
- **Tasks:** Remove irrelevant data, and create new meaningful attributes.
- **Example:** Calculate ratios or differences between existing attributes, such as customer age and spending amount.

5. Model Development and Validation

- **Goal:** Develop and test predictive models.
- **Tasks:** Split the data into training and test sets.
- **Training Set:** Used to build the model.
- **Test Set:** Used to validate and check the accuracy of the model.
- **Example:** Developing a model to predict customer churn and testing its accuracy.

6. Prediction and Interpretation

- **Goal:** Use the best model for making predictions and gaining insights.
- **Tasks:** Apply the model in decision-making processes.

- **Example:** Predict future customer behavior based on the model.
- **Feedback:** The process may involve returning to earlier phases based on results.

Key Roles in the Process

- **Domain Expert:** Defines the problem and provides insights during analysis.
- **Information Systems Expert:** Supervises access to data sources.
- **Mathematical and Statistical Expert:** Conducts data analysis and model development.

Conclusion

The data mining process is a step-by-step approach that requires collaboration between different experts to extract meaningful patterns, develop predictive models, and make data-driven decisions.

Analysis methodologies:

Supervised vs. Unsupervised Learning in Data Mining

Data mining can be divided into **supervised** and **unsupervised** learning, depending on whether a target variable exists.

1. Supervised Learning

- **Target Variable:** A specific outcome is predicted.
- **Goal:** Predict or interpret the target attribute.
- **Examples:**
 - **Classification:** Predicting if a customer will churn (active or churned).

- **Regression:** Predicting future sales based on price and promotions.

2. Unsupervised Learning

- **No Target Variable:** Focuses on discovering patterns.
- **Goal:** Find similarities, clusters, and patterns in data.
- **Examples:**
 - **Clustering:** Grouping customers based on similar buying behaviors.
 - **Association Rules:** Finding products often bought together (e.g., bread and butter).

3. Seven Basic Data Mining Tasks

- **Characterization & Discrimination:** Comparing data distributions to find differences.
- **Classification:** Predicting a target class (e.g., customer loyalty).
- **Regression:** Predicting a continuous value (e.g., sales).
- **Time Series Analysis:** Predicting data that changes over time (e.g., stock prices).
- **Association Rules:** Finding relationships between items (e.g., market basket analysis).
- **Clustering:** Grouping similar data points without pre-defined labels.
- **Description & Visualization:** Summarizing data with charts for easy understanding.

Conclusion

- **Supervised learning** predicts outcomes, while **unsupervised learning** discovers patterns.
 - **Data mining tasks** are used to **classify, predict, cluster, and describe** data for better decision-making.
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Data Preparation:

Creating High-Quality Data for Business Intelligence

To make accurate decisions, **business intelligence systems** need **reliable data**. However, data from primary sources can have **anomalies** that need correction.

1. Key Activities for High-Quality Data

- **Data Validation:** Identify and remove errors and inconsistencies in the data.
- **Data Integration & Transformation:** Combine data from different sources and adjust it to improve accuracy and efficiency for analysis.
- **Data Size Reduction & Discretization:** Reduce the number of attributes and records while keeping the data informative.

2. Importance

- These steps ensure the dataset is **clean and ready** for business intelligence or data mining analysis.

Example:

- A company collects customer purchase data from multiple sources. Data validation removes incorrect entries (e.g., duplicate records), and transformation combines the data into a unified format. Size reduction simplifies the dataset without losing key insights.

Conclusion

- **Reliable, clean data** is essential for effective business intelligence and decision-making.
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Data Validation :

Data Quality Issues and Techniques for Correction

Data quality can be affected by **incompleteness**, **noise**, and **inconsistency**, which need correction for accurate analysis.

1. Incompleteness (Missing Data)

- **Cause:** Missing values may occur due to incomplete records, faulty devices, or manual errors.
- **Correction Techniques:**
 - **Elimination:** Remove records with missing target attributes (but may lose valuable data).
 - **Inspection:** Experts manually check missing values and suggest replacements (time-consuming but accurate).
 - **Identification:** Assign a placeholder (e.g., -1) to missing data.
 - **Substitution:** Replace missing values with:
 - **Mean/Mode:** Average value for numerical data.
 - **Regression Models:** Use predictions to fill in missing values.

Example: If a customer's purchase amount is missing, it may be replaced by the average purchase amount of similar customers.

2. Noise (Erroneous Data/Outliers)

- **Cause:** Random errors, faulty measurements, or inconsistent data entry.
- **Identification Techniques:**
 - **Statistical Dispersion:** Identify outliers using the mean and standard deviation. Values far from the mean are considered outliers.
 - **Clustering:** Group similar records together; values that don't fit any group are considered outliers.
 - **Distance-based Methods:** Identify outliers based on how far they are from other data points.

Example: A customer who spends \$10,000 in a month when most customers spend \$100 could be an outlier (fraudulent or erroneous data).

- **Correction Methods:**
 - **Regularization:** Predict and replace outliers using regression models or use clustering to find normal values.

3. Inconsistency (Discrepancies in Data)

- **Cause:** Changes in data formats or coding systems over time.
- **Correction:** Update records to match the current coding system. For example, change old product codes to new ones after a system update.

Conclusion

Data validation techniques like **elimination**, **inspection**, **substitution**, and **regularization** help correct data issues, ensuring better analysis and decision-making. These techniques ensure data quality in business intelligence and data mining.

Data transformation :

Data transformation is crucial in data mining to improve the accuracy of learning models. Common transformations include **standardization** and **feature extraction**.

1. Standardization (Normalization)

Standardization adjusts the data to a common scale to make it easier for learning models to process.

- **Decimal Scaling:**

- Transforms data by shifting the decimal point to the left.
- Formula: $x'_{ij} = \frac{x_{ij}}{10^h}$
- **Example:** Shifting the decimal point so that values fall within the range $[-1, 1]$.

- **Min-Max Normalization:**

- Rescales data to a specific range, typically $[-1, 1]$.
- Formula:
$$x'_{ij} = \frac{x_{ij} - \min_j}{\max_j - \min_j}$$
- **Example:** Adjusting customer spending data so the minimum value is -1 and the maximum is 1.

- **Z-Index Standardization:**

- Converts data to have a mean of 0 and a standard deviation of 1.
- Formula:
$$x'_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$
- **Example:** If customer spending data follows a normal distribution, the transformed values will typically range from -3 to 3.

2. Feature Extraction

Feature extraction creates new attributes by transforming existing ones, allowing models to capture more complex patterns.

- **Simple Example:**

- If customer spending is recorded over time, **new variables** might be created to capture **trends** (e.g., spending growth or decrease between periods).

- **Complex Transformations:**

- Techniques like **Fourier transforms**, **wavelets**, or **kernel functions** are used for more advanced feature extraction, often in fields like image recognition or time-series analysis.
- **Example:** In **image recognition**, new variables may summarize pixel data to detect patterns like **faces**.

Conclusion

- **Standardization** adjusts data for easier comparison, while **feature extraction** creates new variables to better capture patterns. Both are essential for improving model accuracy in data mining.
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Data Reduction :

Data reduction simplifies datasets for better efficiency, accuracy, and simplicity. Key methods include **sampling**, **feature selection**, **PCA**, and **data discretization**.

1. Sampling

- **Purpose:** Select a smaller, representative part of the data.
- **Example:** Analyzing 1,000 customers instead of 100,000 to save time.
- **Advantage:** Saves time and resources, still provides accurate results.

2. Feature Selection

- **Purpose:** Choose the most important features, remove irrelevant ones.
- **Example:** Keep "income" and "purchase history" for predicting sales, remove "gender".
- **Advantage:** Reduces complexity and improves model performance.

3. Principal Component Analysis (PCA)

- **Purpose:** Reduce the number of features by combining them into principal components.
- **Example:** Reduce 10 features to 2 or 3 new components, capturing most important data.
- **Advantage:** Simplifies data while keeping key information.

4. Data Discretization

- **Purpose:** Convert continuous data into discrete categories.
- **Example:** Group ages into ranges like 0-18, 19-35, etc.
- **Advantage:** Makes continuous data easier to analyze and model.

Conclusion

These techniques **simplify data, reduce computational effort**, and improve **model performance** while preserving important information.